

Task 3 — Network Reconnaissance

Scanning my own lab network with Nmap

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Network	VMware NAT subnet — 192.168.181.0/24 (local lab only)
Date	26 August 2026

1. Objective

Scan a local network to find out what's actually alive on it, then dig into one host specifically — open ports, running services, and OS — using Nmap. Used my own VMware lab network (Kali + the Ubuntu Server from Task 2), nothing external.

2. Tools Used

- Nmap — host discovery, port scanning, service and OS detection

3. Methodology

First swept the whole subnet to see which IPs were actually live, without touching any ports yet — just answering 'who's here.' Once I had a list, I picked out my Ubuntu Server (from Task 2) as the specific target and ran a full scan against it: every port, service versions, and an OS guess, all in one pass.

4. Steps Performed

Found my own subnet:

- `ip a` → 192.168.181.131/24 (my Kali VM)

Host discovery sweep:

- `sudo nmap -sn 192.168.181.0/24`
- Came back with 5 live hosts on the subnet

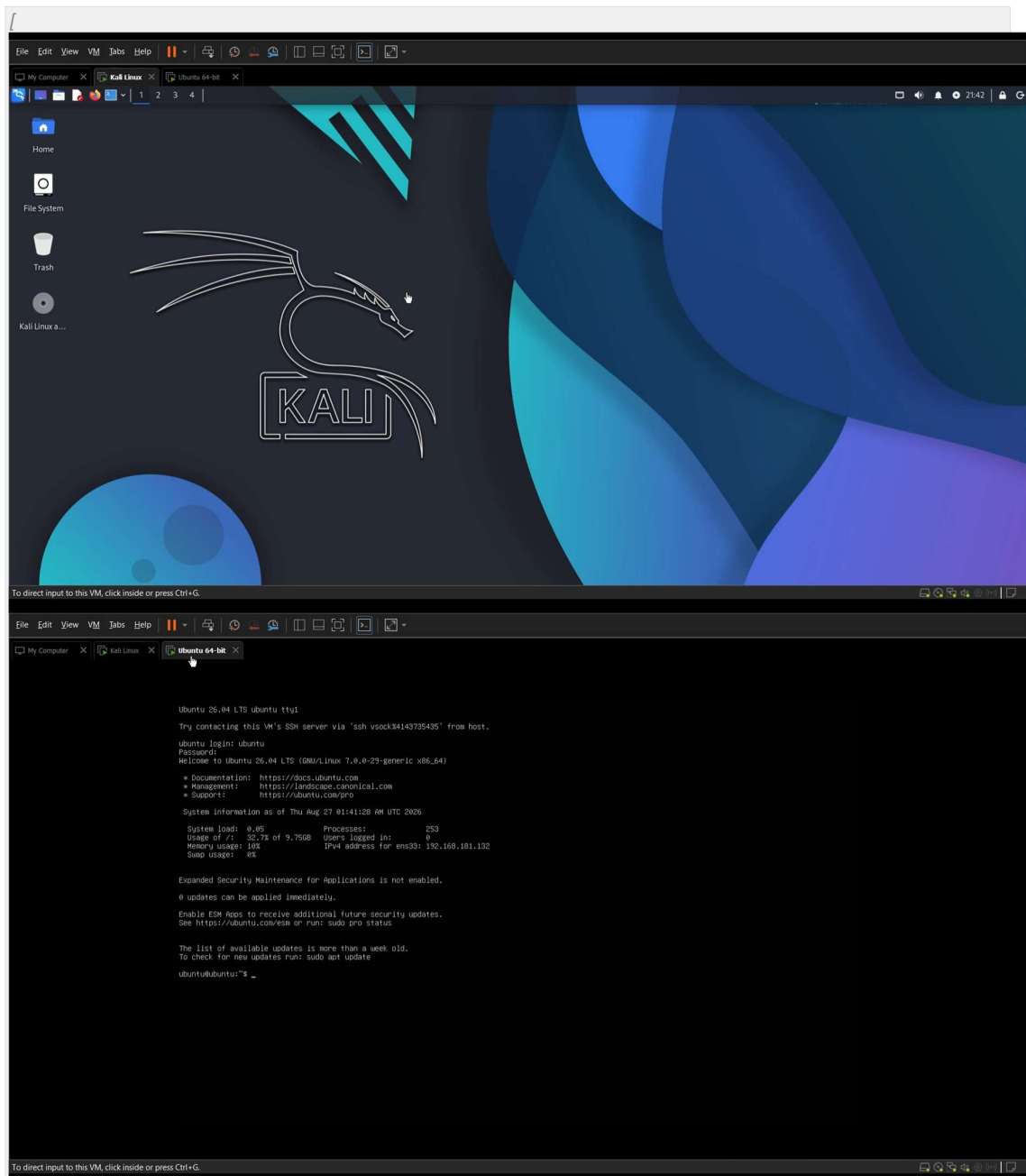
Identified the target:

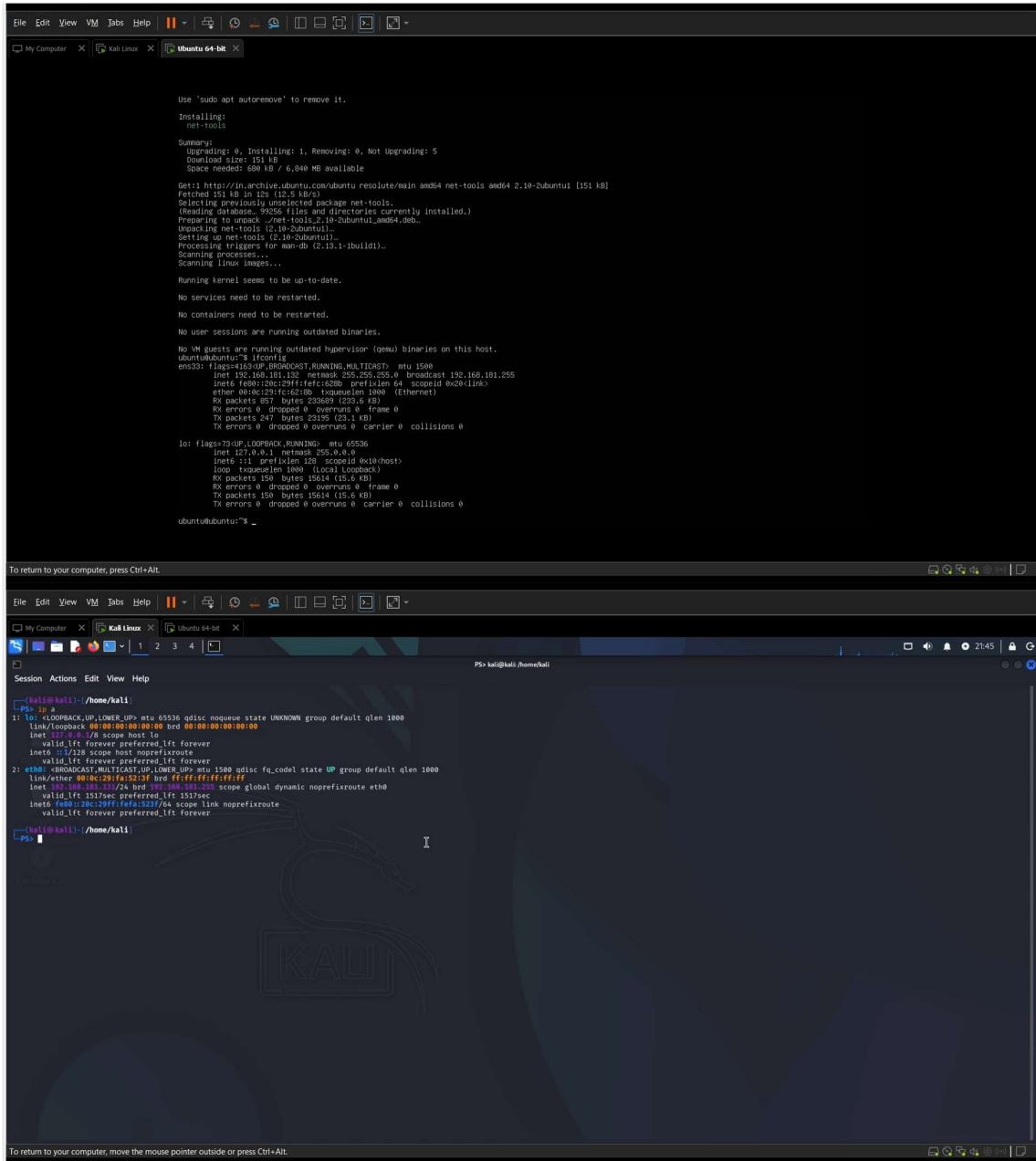
- 192.168.181.132 was the one unaccounted-for device — confirmed as my Ubuntu Server VM

Full scan against the target:

- `sudo nmap -sS -sV -O -p- 192.168.181.132 -`
- Took about 108 seconds to fully complete across all 65,535 ports

5. Screenshots





6. Findings

Hosts discovered on the subnet:

IP	What it is
192.168.181.1	VMware virtual gateway
192.168.181.2	VMware NAT/DHCP service
192.168.181.131	My Kali VM (scanning machine)
192.168.181.132	Ubuntu Server VM — the actual target, from Task 2
192.168.181.254	VMware internal service

Open ports and services on the target (192.168.181.132):

Port	Service	Notes
22/tcp	OpenSSH 10.2p1 (Ubuntu)	Only open port on the whole host — everything else is filtered by UFW

OS detection came back inconclusive — Nmap gave a wide list of low-confidence guesses rather than one clear answer, and said outright the results might be unreliable.

7. Risk Analysis

Honestly, this scan is a good outcome rather than a list of problems. Only one port open (SSH), everything else showing as filtered instead of just closed — that's UFW from Task 2 actively doing its job, not letting Nmap even get a clean response from anything else. The vague OS guess is actually a side effect of that same firewall: Nmap fingerprints OS by comparing responses from an open port against a closed one, and since everything except SSH is filtered (not closed), it didn't have clean signal to work with.

If this had come back with five or six open ports and a confident OS match, that would've been the concerning result — more open ports means more to attack, and a confirmed OS/version gives an attacker a head start on picking known exploits.

8. Recommendations

- Keep the current UFW rule set — one open port (SSH) is close to ideal for a server that only needs remote admin access
- Periodically re-run this same scan after any config changes, to catch anything that accidentally opens up
- If this server ever needs another service (web, database, etc.), open only that specific port rather than loosening the firewall broadly
- Consider disabling ping/ICMP responses if even the discovery sweep should be harder to perform from outside the trusted network

9. Conclusion

Swept the lab subnet, found five live hosts, and ran a full port/service/OS scan against the Ubuntu Server from Task 2. The result directly reflects that earlier hardening work — one open port, everything else filtered, and an OS fingerprint that Nmap couldn't pin down confidently. This task tied straight back into Task 2 rather than being a separate exercise, which made it easier to actually understand what the scan was showing instead of just reading output.